Searching for Patterns in the Optimal Controls which Prepare the Ground State of a Superconducting Device

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Numerical solutions to complex many-body models are severely limited by classical computational power. Using controllable quantum devices may help with the simulation of these models. One approach, known as the variational quantum algorithm, is based on tailoring evolution to prepare the a priori unknown ground states of model Hamiltonians. In this work, we address the challenge of finding patterns in optimal variational protocols, where an optimal protocol minimizes the total time it takes to get from the ground state of one device to the ground state of another target device. Using a classical computer, we simulate the evolution of superconducting gmon qubits and use Monte-Carlo (MC) simulations to find this optimal protocol for finite systems sizes. The MC method finds optimal protocols that are roughly bang-bang, as predicted by theory. We compare the MC method with an adiabatic method and show that it reaches the ground state in shorter times. Further research will identify patterns in these protocols that can be extrapolated to larger system sizes in order to inform efficient large-scale simulations on the quantum device.

I. INTRO

There are many techniques for finding the ground-states of complex many body models (WKB, adiabatic, references here). A general approach involves starting from an easy-to-prepare initial state and evolving into the desired ground state. This evolution is carried out by a unitary operator which is generated by the controls, u, of the target Hamiltonian.

$$\begin{aligned}\psi_{target} &= U(u(t))\psi_{initial} \\ &= e^{-itH(u(t))}\psi_{initial}\end{aligned}$$

A common protocol, the value of the controls over time, follows the adiabatic method. This method starts with the controls equal to the Hamiltonian that generates the initial state and linearly evolves into the controls of the target state. If done sufficiently slowly in the absence of a vanishing spectral gap, this approach should effectively prepare the desired ground state. However, this is limits the targets we can reach due to the spectral gap requirement, and even in the case of a non zero gap, the required times makes some targets impractical.

A promising alternative to the adiabatic method relies on a result from optimal control, Pontryagin's minimum principle. This minimum principle implies that, if a given set of conditions are met, the optimal path has controls which take on either their maximum or minimum value at any given time – a bang-bang protocol. We explore this by searching for the optimal protocols which prepare a desired target state using three different type of Monte-Carlo (MC) simulations on a classical compute. The first Monte-Carlo Brute-Force (MCBF) method does not assume bang-bang protocols, but still converges to them. The second, Monte-Carlo Bang-Bang (MCBB) assumes bang-bang protocols and outperforms the MCBF. The third, Monte-Carlo Discrete Bang (MCDB) assumes bang-bang as well but sacrifices a small amount of accuracy in exchange for computational efficiency and performs almost as well as the MCBB. For a fixed initial and target, we find almost identical protocols for the 3 approaches. We compare these to the adiabatic method and the 3 MC methods, which have comparatively identical performance, all outperforming the adiabatic.

Generating the unitary matrix involved in this evolution takes $\mathcal{O}(N^3)$, where N is the dimension of the Hamiltonian. This computational demand limits the ability to find these optimal protocols for larger system sizes. One way to get around this is to do the evolution on a tuneable superconducting device. We can achieve the desired optimal protocol by iteratively changing an ansatz protocol and minimizing the expectation value of the evolved state $\langle\psi_t|H_t|\psi_t\rangle$ until this reaches a plateau which corresponds to the ground state energy of the target Hamiltonian. This implies successful evolution into the desired target state.

While this approach avoids the computational complexity through direct measurement, finding the optimal protocol could still be difficult due to the complexity of the control-space and the number of iterations required to find a global minimum of the expectation value. Our work aims to mitigate this issue by finding patterns in the classically-obtained protocols for smaller system sizes which we hope will extrapolate to larger system sizes and inform an efficient search for the optimal protocol on the superconducting device. With this extrapolation, honing in on the optimal protocol involves significantly fewer iterations.

II. THE XXZ MODEL

Although it is possible to simulate models which are distinct from the quantum device through a mapping, we choose to simulate the XXZ model which governs the su-

perconducting device. The Hamiltonian for this follows:

$$H_{device} = \sum_{<ij>} j_{ij}(\sigma_i^x \sigma_j^x + \sigma_j^y \sigma_i^y) + k_{ij}\sigma_i^z \sigma_j^z + \sum_i b_i \sigma_i^z$$

We simplify the model by only considering uniform sites, so $J = j_{ij}$, $K = k_{ij}$, $B = b_i$ $\forall$ sites i,j on our lattice. With this simplification, the value of B contributes an unimportant phase factor during the evolution so we drop the term. Simulating this 1D model on a square lattice is possible by snaking the 1D through the. Say more here.

For a square lattice with M sites and C occupants, the dimension of the Hamiltonian becomes $\binom{M}{C}$. We find that we require a dimension of 4000 or smaller to make finding the optimal protocols computationally feasible with the computing power we have access to. This means we're able to explore all occupancies in a 2x2 and 3x3 lattice classically, along with some small occupancies for 4x4 and 5x5 systems. We also skip systems with 0 and 1 occupants as there isn't any interesting interaction going on (phrase this differently). Given a lattice with C occupants, we find that the same lattice with N-C occupants has the exact same evolution with different energy scales. This symmetry allows us to ignore occupancies with more than half filling.

III. DISTANCE MEASUREMENT AND ENERGY SCALE INDEPENDENCE

Throughout the Monte-Carlo simulation, our goal is to minimize a cost function in either energy space or state overlap space which are calculated as:

$$C(\psi_t)_{energy} \equiv \langle \psi_t | H_t | \psi_t \rangle$$
$$C(\psi_t)_{state} \equiv 1 - \langle \psi_t | \psi_{initial} \rangle^2$$

where ψ_t is the evolved state. Upon successfuly evolving into the target state, C_{state} vanishes and $C_{energy} = E_o$. For longer timescales, there are many different paths which evolve into the desired target state and getting the exact minimum total time where we have perfect convergence is computationally expensive, so we choose to find the optimal total time which evolves just short of the target state ($C_{state} = 0$ or $C_{energy} - E_o = 0$). In order to stop just short of the target state in a meaningful way, we introduce a normalized 'distance' measurement which represents the distance left until we reach the target state. This is defined in energy and state space respectively as:

$$D(\psi_t)_{energy} \equiv \frac{\langle \psi_t | H_t | \psi_t \rangle - E_o}{\langle \psi_{initial} | H_t | \psi_{initial} \rangle - E_0}$$
$$D(\psi_t)_{state} \equiv \frac{1 - \langle \psi_t | \psi_{target} \rangle^2}{1 - \langle \psi_t | \psi_{initial} \rangle^2}$$

It's worth noting that these are both independent of the energy scale, and only dependent on the ratio of j and k due to the fact that the states are normalized and

$$D(\psi_t)_{energy} = \frac{\alpha}{\alpha} \frac{\langle \psi_t | H_t | \psi_t \rangle - \langle \psi_{target} | H_t | \psi_{target} \rangle}{\langle \psi_{initial} | H_t | \psi_{initial} \rangle - \langle \psi_{target} | H_t | \psi_{target} \rangle}$$
$$= \frac{\langle \psi_t | \alpha H_t | \psi_t \rangle - \langle \psi_{target} | \alpha H_t | \psi_{target} \rangle}{\langle \psi_{initial} | \alpha H_t | \psi_{initial} \rangle - \langle \psi_{target} | \alpha H_t | \psi_{target} \rangle}$$

With this distance measurement, we stop our Monte-Carlo simulations when $D(\psi_t) \leq 0.01$ and call the total time involved in the evolution of ψ_t our critical time, $\tau_{critical}$. We can approximate the exact critical time by doing a low order polynomial fit to the distance as a function of total time and interpolate the time where $D(\psi_t) = 0$.

IV. THE ADIABATIC METHOD

Evolving from an initial to a target state is usually carried out by the adiabatic method. This starts with the controls equal to the Hamiltonian that generates the initial state and linearly evolves into the controls of the target state. If done sufficiently slowly in the absence of a vanishing spectral gap, this approach should effectively prepare the desired ground state.

In order to approximate this, we discretize time and do a rough approximation to the linear protocol. For a fixed initial time, we evolve the initial state according to this linear protocol, and then increment the total time until we can successfully evolve into the target state.

V. BRUTE FORCE MONTE-CARLO METHOD

Although the ideal protocol is a continuous function, our first method dubbed the Brute-Force Monte-Carlo (MCBF) approximates this by discretizing time and allowing the protocols to take on any value within the bounds of our parameters, in this case [0,1]. We select a total time, τ and a fixed number of intervals N. The evolution of the state then follows

$$\psi_{evolved} = U(\{J_i\}, \{K_i\})\psi_{initial}$$
$$= \prod_i^N e^{-i\frac{\tau}{N}H(J_i,K_i)}\psi_{initial}$$

We pick a random initial protocol, sufficiently small initial total time τ_o and pseudo-temperature T, evolve the initial state, and calculate the expectation value of the evolved state. We then follow this Simulated-Annealing procedure:

1. Randomly select an interval, change the value of the protocol by some small random amount at that interval

2. Repeat the evolution, measure the new expectation value E_{new}

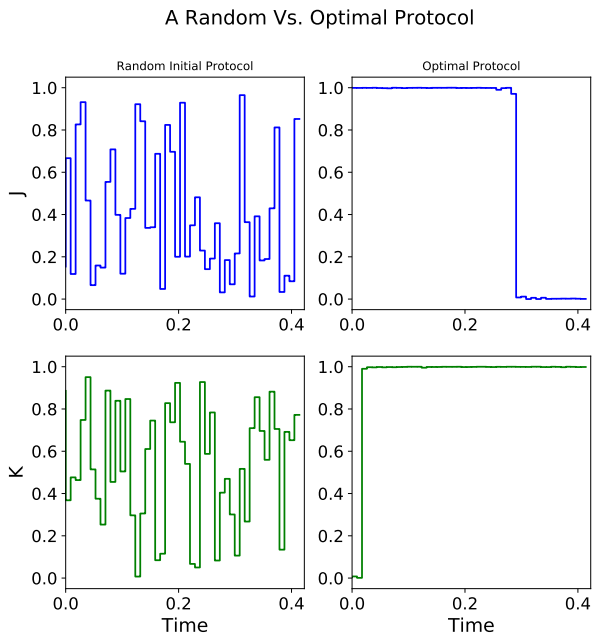

FIG. 1. LHS showing a random initial protocol for J and K and RHS showing the optimal post-MCBF protocol

3. If this value is smaller than the old, keep the change. Otherwise, keep the change with some probability $e^{-\frac{E_{new}-E_{old}}{T}}$.

4. Repeat 1-3 for S sweeps, then decrease T by some small amount

5. Repeat 1-4 L times, then increase the total time

6. Repeat 1-5 Until the expectation value equals the ground state energy of the target Hamiltonian

The number of sweeps S and temperature-decay iterations L should be chosen to get sufficiently close to the optimal protocol for each τ. To have confidence that you are in fact reaching the optimal protocol for each τ, this process can be done for multiple seeds which create different initial protocols and different changes throughout the process, but converge on the same protocol. We have a pseudo-temperature T to allow for change which take us further away from the ground state in order to prevent getting stuck in some local minimum. This process can also be repeated for a different number of total steps N until an increase in N doesn't provide much more convergence. For our case, N = 20 was typically sufficient.

After doing this for a variety of initial and target Hamiltonians, a pattern emerges. The protocols collapse into bang-bang protocols, approaching either the maximum or the minimum value (1 or 0) shown in Figure 1.

Comparing this to the adiabtic approach, we find significantly better performance as shown in Figure 2

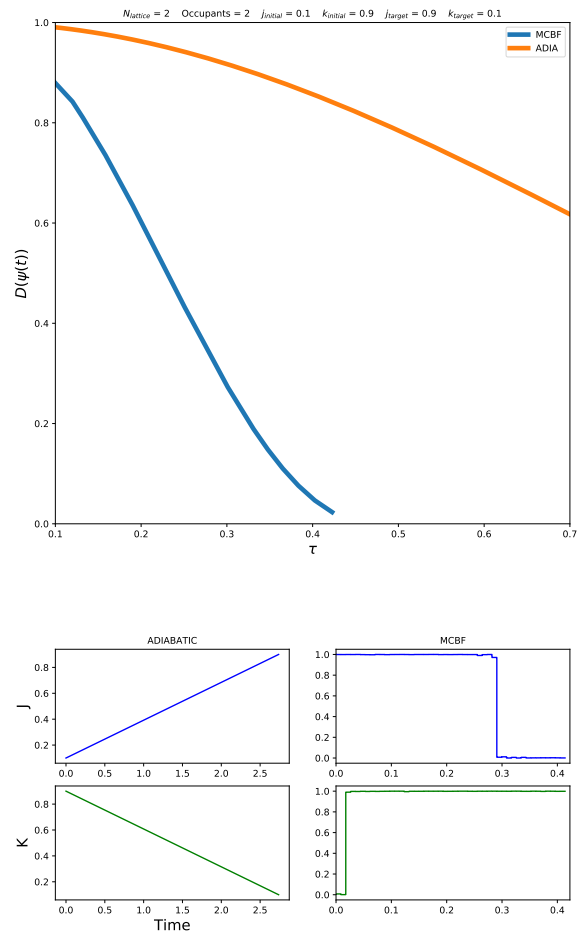

FIG. 2. **Top:** An example of the distance vs τ for the two methods. The MCBF achieves the ground state in a much shorter time. **Bottom:** The final protocols for the two methods

VI. PONTRYAGIN'S MINIMUM PRINCIPLE

VII. BANG-BANG MONTE-CARLO TECHNIQUES

With the MCBF converging on bang-bang protocols, we choose to run a secondary Bang-Bang Monte-Carlo (MCBB) algorithm. For the MCBB, the variatonal parameter is no longer the value of the intervals, but the time that the protocol jumps between 1 and 0. The MCBF provides a rough number of jumps but we overestimate the number of jumps in order an allow the excess jumps to 'collapse'. For identical τ's, the MCBB technique always performed as well or better than the MCBF, and always collapsed to a protocol that was similar in shape to the MCBF protocol. The improvement comes from avoiding the discrete MCBF time intervals and allowing J and K to reach the true minimum/maximum. This confirms the bang-bang protocols predicted by Pon-

tryagin's minimum principle.

VIII. OPTIMIZATION TRICKS

A. Discrete-Bang Monte-Carlo

Moving from the MCBF to the MCBB improved converge at each τ step, but also saves computing time. The MCBF allows us to get a rough estimate for the number of square pulses. For N square pulses there 4*N variatonal parameters in the MCBB where a factor of 2 comes from us requiring a turn-on and turn-off time for each square pulse, and another factor of 2 coming from the fact that we have 2 different J and K parameters. For smaller dimensions, the MCBF rarely found more than 2 jumps. This means that we only have 8 timestamps which can vary. Further, the number of sweeps required drops because we have fewer variational parameters. This gives the MCBB a significant computing advantage over the MCBF.

However, even the improvements in the MCBB aren't enough to make some system sizes tractable. The most computationally expensive part of the entire MC process comes from the exponentiation of the hamiltonian which generates the unitary operator. One can use a Pade approximation rather than diagonalizing to mitigate some of this complexity but the process is still $\mathcal{O}(N^3)$ where N is the dimension of the Hamiltonian. In order to get around this we try a Monte-Carlo Discrete-Bang approach which goes back to fixed discrete time intervals but only allows each interval to take on the value of 1 or 0. This means we have 3 possible values of the Hamiltonian: H_{10}, H_{01}, H_{11} which correspond to the 3 meaningful combinations of J and K. With only 3 possible exponentiations that could take place, we pre-compile the Unitary Operators. We exclude H_{00} because ($e^{aH_{00}}$ generates the identity matrix which is wasted time during the evolution). By precompiling, the evolution is reduced from $\mathcal{O}(N^3)$ to $\mathcal{O}(N^2)$ matrix-vector multiplication.

Applying this to the same systems yields results that outperform the MCBF and perform almost as well as the MCBB but take significantly less computing time for larger system sizes. With this, we find it feasible to push into regions with Hiblert spaces of dimension up to 4000. A comparison of the 3 MC simulations and the adiabatic approach are shown in Figure 3. The MCDB approach has 256 total steps per evolution gauntlet for a fixed time where the max number of steps is 128, at a cost of $\mathcal{O}(N^2)$. The MCBB has roughly 8 steps at a cost of $\mathcal{O}(N^3)$. This implies the MCDB approach beats the MCBB when $256N^3 = 8N^2$, or $N = 32$. All meaningful system sizes, with the exception of a 2 by 2 lattice with 2 occupants, have dimensions greater than this so the MCDB approach always outperforms.

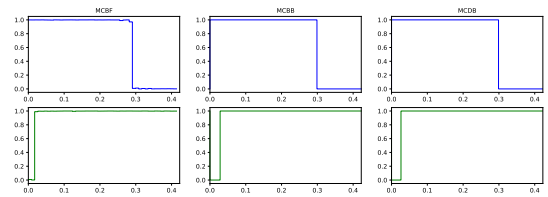

FIG. 3. The final protocols for the 3 different MC methods. The differences between the MCBB and the MCBF for a fixed total time is < 0.001, with the MCDB performing between the two others.

B. Saving Each State During Evolution

Another optimization shortcut relies on the fact that a change in the protocol at a given interval does not change the state leading up to that change. This means that we can save the state at every step in the evolution and only 'continue' the evolution from the change onward. Since the index that recieves the change is uniformly distributed across all possible steps, this results in only $\langle \frac{N}{2} \rangle$ steps on average.

C. Adaptive Step Size

Too few steps makes the evolution coarse and restricts the time that these jumps can occur. Too many steps makes the MCDB computationaly expensive and makes it difficult to find the optimal protocol with so many indices to choose from. In order to get around this, we add another loop to our MCDB which, for a fixed total time, starts with N=4 steps, runs the MCDB and finds the optimal protocol, then doubles the number of steps iteratively until N=128 or 256. Each time the number of steps are increased, the initial protocol for the next MCDB run is the optimal protocol for the previous.

D. Penalizing Fictitous Jumps

To save steps during the Monte-Carlo, we bias our index selection towards points near a jump. It's unlikely that a single time-step getting changed in the middle of a plateau is going to get us closer to the target state. Progress is more likely to be made by slightly shifing the time a jump occurs. To carry out this bias we add a 'reroll' if an index is selected which has identical neighbors.

IX. OPTIMAL PROTOCOL'S ENERGY SCALE INDEPENDENCE

All simulations searched for optimal protocols by trying to minimize one of the two cost functions defined in

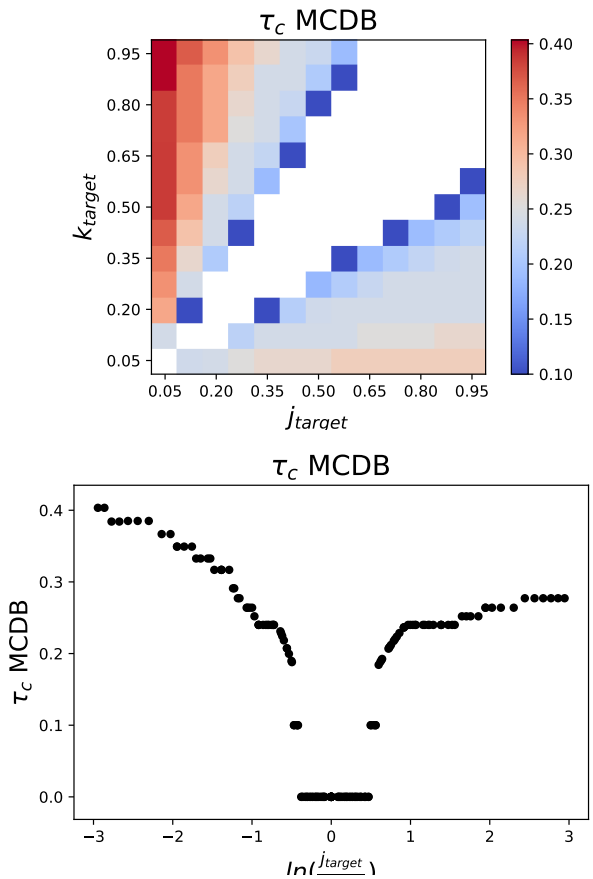

FIG. 4. **Top:** $\tau_{extrapolated}$ for $J_i = K_i = 0.05$ over a 13x13 grid of targets within the parameters range. White space indicates no data due to the fact that the initial and target hamitlonian were already so close. For a fixed ratio, total time required was exactly the same. **Bottom:**The same data collapsed down to only being depdent on the ratio of the targets. For a fixed ratio, there is only 1 visible total time even though there are many different energy scales for that ratio.

section . In the case of the energy dependent cost function, we terminate when $D \leq 0.01$, which is shown by equation to be independent of the energy scale. For the state dependent cost function, the normalization of the states removes any eneregy scaling. For these reasons, the protocols that prepared our target state were only dependent on the ratios of the initial and target. This is highlight by Figure 4, which shows total time for an MCDB run over a variety of targets with $J_i = K_i = 0.05$. For targets with similar ratios, we get the exact same result. When comparing the total time as a function of the ratio, all point for the same ratio collapse down into a single point. This is true for other measures, as well as the shape of the protocol.

Although the energy scale is necessary for simulation on a quantum chip (energy expeecation value is the only cost function we can physically measure), energy distance becomes redundant for our simulation. This allows for a classical computing shortcut - only run the simulation over a variety of unique ratios.

X. CHARACTERIZING CORRELATIONS BETWEEN BANG-BANG PROTOCOLS

The correlation function is defined as:

$$C(\{a_i\},\{b_i\}) \equiv \frac{1}{N}\sum_i^N [(a_i + b_i + 1)\text{mod}2]$$

and measures the fractional overlap of two discrete-bang protocols with the same number of steps, where the values of a_i, b_i are only allowed to be 1 or 0. $C(\{a_i\},\{b_i\}) = 1$ implies identical normalized protocols, and $C(\{a_i\},\{b_i\}) = 0$ implies perfectly anti-correlated protocols. This measure implies that perfectly anti-correlated protocols are are only possible when the jumps occur at the same time. Accounting for the fact that the expected output of C varies based on the number of jumps in $\{a_i\}$ and $\{b_i\}$, we introduce the modified correlation function:

$$C_m(\{a_i\},\{b_i\}) \equiv C(\{a_i\},\{b_i\}) - \langle C(N_{a-jumps}, M_{b-jumps})\rangle$$

which calculates the difference between the protocol overlap and the expected protocol overlap given the number of jumps in each of the protocols. $C(\{a_i\},\{b_i\}) \epsilon [0,1]$ and $\langle C(a_{jumps}, b_{jumps})\rangle \epsilon [0.5, 0.6]$ implies $C(\{a_i\},\{b_i\}) \epsilon [-0.6, 0.5]$ where a positive[negative] value shows correlation[anti-correlation] and 0 means no correlation. The expected values of $\langle C(N_{a-jumps}, M_{b-jumps})\rangle$ are shown in Figure 5 and were generated using a numerical simulation with 1,000,000 random samples each.

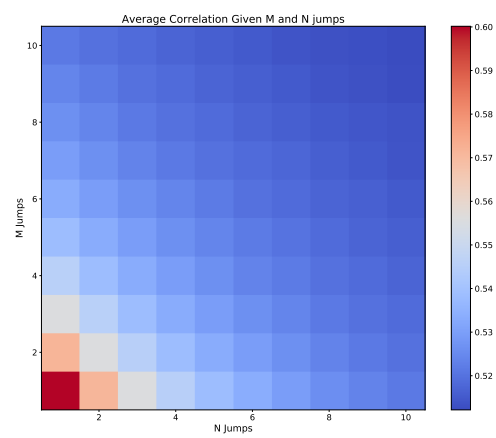

FIG. 5. Each expectation value calculated using a numerical simulation that does 1,000,000 comparisons for randomly distributed protocols with N and M jumps

XI. SYNTHESIS

A. Total Time Comparisons between Adiabatic and Optimal Protocols

In every case, we found the optimal protocol to be bang-bang in nature. This always outperformed the adiabatic approach, with the ratios in total times required increase with dimension.

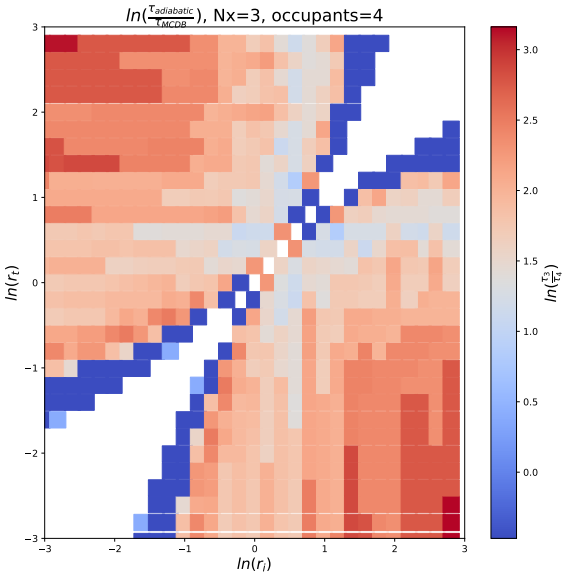

FIG. 6. A representative sample of the time discrepancy between the optimal protocol from the total time required for adibatic evolution vs MCDB total time. Plotting the $ln(\frac{\tau_{adiabatic}}{\tau_{MCDB}})$ against the initial and target ratio. All combinations, with the exception of the blue spots where $r_i \approx r_t$, take significantly longer for the adiabatic method

B. Increasing N can Decrease τ_{avg}

Although systems with a larger Hilbert Space lead to an increase in classical computing time, we sometimes find the opposite to be true for the optimal protocols among different system sizes. As shown in Figure 7, for $J_i = K_i$, and $J_t \leq K_t$, $\tau_{extrapolated}$ is larger for a 3x3 system with 3 occupants than for a 3x3 system with 4 occupants. Similar behavior shows up for other target and initial combinations. Figure 8 shows two instances of the average total time required to achieve the target state being lower in a more complex system – from 3x3 3 occupants to 3x3 4 occupants and 5x5 2 occupants to 5x5 3 occupants.

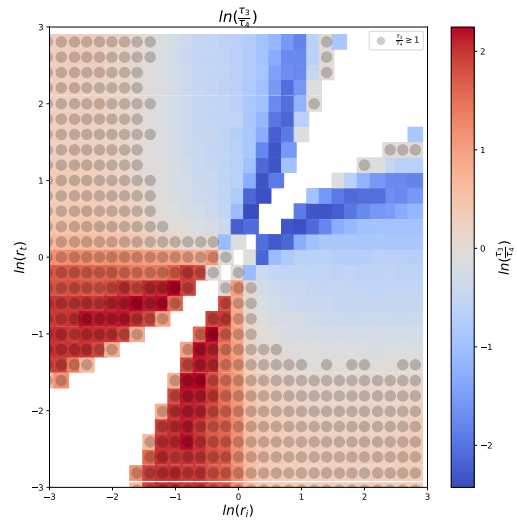

FIG. 7. The ratios of $\tau_{extrapolated}$ between 3 and 4 occupants across all combinations of r_i, r_t. For $r_i \leq 1, r_t \leq 1$, most optimal protocols for 3 occupants had total times which were greater than 4 occupants. Points with dots show where this is the case.

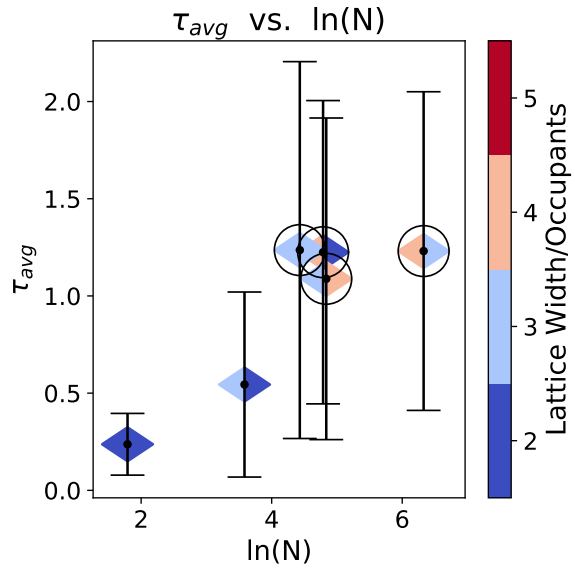

FIG. 8. The average of $\tau_{extrapolted}$ vs the dimension of the hamiltonian. Circled points highlight that an increase in occupancy for a fixed lattice size can sometimes decrease the average critical time – Increasing the dimension can sometimes decrease quantum simulation time

C. Symmetries about the diagonal in r_i vs r_t Figures

For a fixed initial and target state, evolution is determined by:

$$\begin{aligned}\psi_{r_t} &= U(u(t))\psi_{r_i} \\ &= e^{-itH(u(t))}\psi_{r_i}\end{aligned}$$

but if we switch the target and initial, we can do the same evolution with the inverse of the unitary operator. This means our protocol is flipped, but measures such as total time and number of jumps should be symmetric about the diagonal

D. Jump Regimes and the Correlation Function

1. *Continuous Changes in the number of Jumps*

We conjecture that the changes in the number of jumps in r_i, r_t space is continuous ie we don't go from 1 to 3 jumps, but rather 1 to 2 to 3. We see all places with 3 jumps located inside an insland with 2 jumps. For almost

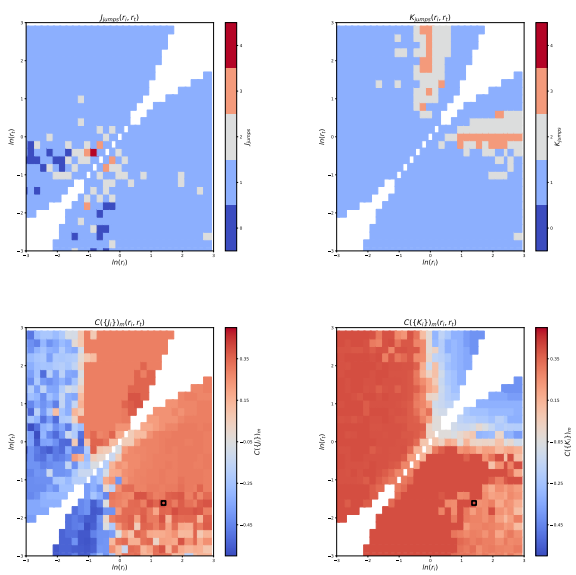

FIG. 9.

all system sizes that we've explored, with the exception of 2x2 2 occupants, a common theme emerges: Most of the optimal protocols have a single jump. However, for $J\{t\}$, when both $r_i < 1$ and $r_t < 1$, J increases to 2 or 3 jumps. For $K\{t\}$, when both $r_i > 1$ and $r_t > 1$, we also see a transition from 1 to 2 or 3 jumps. Additionally, the protocols in the region with more than 1 jump anti correlate with the regions that have a single jump, shown in Figure 9

E. Patterns in Total On Time

We find that for all system sizes, the total time required increases as we leave the diagonal where $ln(r_i) = ln(r_t)$. The increase in total time is roughly proportional to $|ln(\frac{r_i}{r_t})|$, with the proportionality constant typically increasing for system sizes with larger hilbert spaces. For a 2x2 lattice with 2 occupants shown in the LFS of Figure 10, there are no transitions which results in smooth increases in total time. For 3x3 with 4 occupants, there is a transition occurring around the time $ln(r_i) = 0$ or $ln(r_t) = 0$ which results in the square shape shown on the RHS.

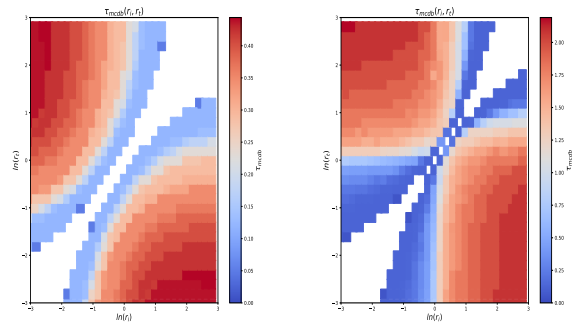

FIG. 10. Two representative figures for the total time required for the optimal protocol. The total time is roughly proportional to $|ln(\frac{r_i}{r_t})|$

F. Fractional On-Time Across Different System Sizes

Another pattern: Letting $p = r_i * r_t$ and starting with $p \approx 0$, the optimal protocols across system sizes have J off a majority of the time, and K on for the entire protocols. As $p- > e^6$ (our arbitrary upper bound that we chose to scan over), the role of J and K switch. J is for the entire optimal protocol, and K is barely turned on at all, shown in Figure 11.

G. Similarities Across System Sizes

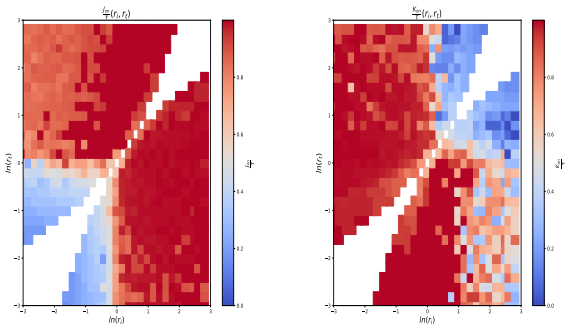

FIG. 11.

H. temporary spacer

I. temporary spacer

J. temporary spacer

K. temporary spacer

L. temporary spacer

M. temporary spacer

N. temporary spacer

O. temporary spacer

P. temporary spacer

Q. temporary spacer

R. temporary spacer

S. temporary spacer

T. temporary spacer

U. temporary spacer

V. temporary spacer